

Topics in Error-Correcting Codes

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Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Logistics

All the logistics for this class can be found on the [course website](#).

Recap

Definition 1. A code $\mathcal{C} \subseteq \{0,1\}^n$ is called a binary *linear code* if it is nonempty and, for all $x, y \in \mathcal{C}$, we have $x \oplus y \in \mathcal{C}$.

Claim 1. For every nontrivial binary linear code $\mathcal{C}$, its distance is equal to the minimum Hamming weight among all nonzero codewords. Namely,

$$\min_{0 \neq c \in \mathcal{C}} \text{wt}(c) = \min_{c_1 \neq c_2 \in \mathcal{C}} \Delta(c_1, c_2)$$

Proof. For any distinct $c_1, c_2 \in \mathcal{C}$, the word $c_1 \oplus c_2$ is a nonzero codeword and

$$\Delta(c_1, c_2) = \text{wt}(c_1 \oplus c_2).$$

Therefore,

$$\min_{0 \neq c \in \mathcal{C}} \text{wt}(c) \leq \min_{c_1 \neq c_2 \in \mathcal{C}} \Delta(c_1, c_2).$$

Conversely, $0^n \in \mathcal{C}$, and for every nonzero $c \in \mathcal{C}$ we have

$$\text{wt}(c) = \Delta(c, 0^n).$$

It follows that

$$\min_{0 \neq c \in \mathcal{C}} \text{wt}(c) \geq \min_{c_1 \neq c_2 \in \mathcal{C}} \Delta(c_1, c_2),$$

which proves the claim. □

Remark. The operation $\oplus$ is simply addition over $\mathbb{F}_2$, the field with elements $\{0, 1\}$. We will often use $\mathbb{F}_2$ to denote the binary alphabet.

Theorem 2 (Hamming Codes). *For every integer $r \geq 2$, there exists a binary $[2^r - 1, 2^r - r - 1, 3]$ linear code.*

Remark. Last class, you saw the construction of the $[7, 4, 3]$ Hamming code, which is the case $r = 3$.

Family of Codes

When we perform an asymptotic study of codes, it makes sense to consider a family of codes and study its asymptotic rate and distance.

Definition 3. Let $\mathcal{C} = \{\mathcal{C}_i\}_{i \in \mathbb{N}}$ be a family of codes, where each $\mathcal{C}_i$ is an $(n_i, k_i, d_i)_{q_i}$ code and $n_i \rightarrow \infty$. We define the rate of this family by

$$R(\mathcal{C}) = \liminf_{i \rightarrow \infty} \frac{k_i}{n_i}$$

and its relative distance by

$$\delta(\mathcal{C}) = \liminf_{i \rightarrow \infty} \frac{d_i}{n_i}.$$

Question: Does there exist a constant q and a family of codes $\mathcal{C}$ such that $q_i = q$ for every $i \in \mathbb{N}$, $R(\mathcal{C}) > 0$, and $\delta(\mathcal{C}) > 0$?

A family satisfying these conditions is called **asymptotically good**.

Spoiler: We will present several families of asymptotically good codes.

A few motivating puzzles

The following puzzle was my first encounter with coding theory, although I did not realize it at the time.

Question 4. How large can a set of strings in $\{0, 1, \dots, 9\}^{10}$ be if every pair of strings differs in at least two digits?

Proof. The maximum is 10^9 . We prove matching upper and lower bounds.

Upper bound: There are at most 10^9 such strings because their length-nine prefixes must be pairwise distinct.

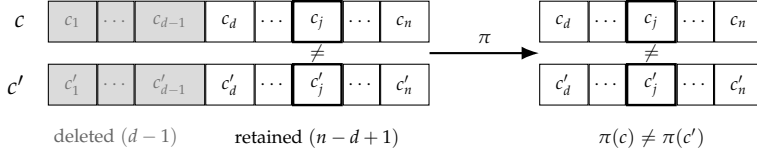
Lower bound: Consider all strings whose digit sum is $0 \pmod{10}$. Every choice of the first nine digits uniquely determines the final digit, so there are 10^9 such strings. Two of them cannot differ in exactly one digit: equality of their digit sums modulo 10 would force the two values in that coordinate to be equal. Thus every pair differs in at least two digits. $\square$

The following bound is due to [Singleton \[1964\]](#).

Theorem 5 (Singleton Bound). *For every $(n, k, d)_q$ code $\mathcal{C} \subseteq \Sigma^n$, where $|\Sigma| = q$, we have*

$$k + d \leq n + 1.$$

Proof. Delete any $d - 1$ coordinates from every codeword. The resulting projection is injective: otherwise, two distinct codewords would differ only in those $d - 1$ coordinates, contradicting the minimum-distance guarantee.



Since there are q^{n-d+1} possible projections,

$$q^k = |\mathcal{C}| \leq q^{n-d+1}.$$

Taking logarithms base q gives the desired inequality. $\square$

Definition 6. A code matching the singleton bound is called an MDS code. Above we constructed a $(10, 9, 2)_{10}$ code.

If $\mathcal{C}$ is an MDS code, then deleting any $d-1$ coordinates gives an injection from $\mathcal{C}$ to Σ^k , where $k = n-d+1$. Since both sets have size q^k , every such projection is a bijection. Thus every codeword is determined completely by any k of its coordinates.

Remark. The construction motivates the idea of *parity-checks* or *sum-checks*! Note that checking membership of a valid phone number in our construction is extremely efficient. You may remember that credit cards in fact use a funkier form of such sumchecks (the Luhn sumcheck) and that's how websites detect efficiently if you entered a valid credit card number.

Question 7. Can you construct $(10, 8, 3)$ codes?

Linear Codes and Finite Fields

More generally, we use the following definition of a linear code.

Definition 8 (Linear Codes). For any prime power q , a code $\mathcal{C} \subseteq \mathbb{F}_q^n$ is linear if $\mathcal{C}$ is a linear subspace of $\mathbb{F}_q^n$.

Remark. A finite field $\mathbb{F}_q$ has q elements, where q is a prime power. In a field, we can add, subtract, and multiply, and we can divide by every nonzero element. For every prime p , the integers modulo p form the finite field $\mathbb{F}_p$.

A nonempty subset $\mathcal{C} \subseteq \mathbb{F}_q^n$ is a linear subspace if

$$c_1, c_2 \in \mathcal{C} \implies c_1 + c_2 \in \mathcal{C} \quad \text{and} \quad c \in \mathcal{C}, \alpha \in \mathbb{F}_q \implies \alpha c \in \mathcal{C}.$$

Definition 9 (Generator matrix). Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a linear code with $k = \dim(\mathcal{C})$. A *generator matrix* for $\mathcal{C}$ is a matrix $G \in \mathbb{F}_q^{k \times n}$ such that

$$\mathcal{C} = \{mG \mid m \in \mathbb{F}_q^k\}.$$

Such a matrix exists by taking any basis of $\mathcal{C}$ as the rows of G .

Observe that any row of the generator matrix is in fact itself always a codeword!

Definition 10. Given a full-rank matrix $G \in \mathbb{F}_q^{k \times n}$, the code generated by G is $\{mG \mid m \in \mathbb{F}_q^k\}$.

Definition 11 (Parity Check matrix). Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a linear code with $k = \dim(\mathcal{C})$. A *parity-check matrix* for $\mathcal{C}$ is a matrix $H \in \mathbb{F}_q^{n \times (n-k)}$ such that

$$\mathcal{C} = \{x \mid x \in \mathbb{F}_q^n, xH = 0\}.$$

In fact taking a full rank matrix H such that $GH = 0^{k \times (n-k)}$ suffices.

q-ary entropy

Definition 12. For integers $n, \ell \geq 0$, let the volume of a Hamming ball of radius ℓ in Σ^n , where $|\Sigma| = q$, be denoted by $\text{Vol}_q(n, \ell)$. Then

$$\text{Vol}_q(n, \ell) = \sum_{j=0}^{\ell} \binom{n}{j} (q-1)^j.$$

Theorem 13 (Hamming Bound). For any $(n, k, d)_q$ code,

$$k \leq n - \log_q(\text{Vol}_q(n, \lfloor (d-1)/2 \rfloor))$$

Gilbert [1952] proved the following bound for general, not necessarily linear, codes.

Theorem 14 (Gilbert-Varshamov bound). For any $1 \leq d \leq n$, there exist $(n, k, d)_q$ codes with

$$k \geq n - \log_q(\text{Vol}_q(n, d-1))$$

Proof. We construct the code greedily:

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 $\mathcal{C} \leftarrow \{0\}$ 
while  $\exists x \in \Sigma^n \setminus \mathcal{C}$  such that  $\Delta(x, c) \geq d$  for every  $c \in \mathcal{C}$  do
   $\mathcal{C} \leftarrow \mathcal{C} \cup \{x\}$ 
return  $\mathcal{C}$ 

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Such an element exists whenever $|\mathcal{C}| \text{Vol}_q(n, d-1) < q^n$. Therefore, when the algorithm stops,

$$q^n \leq |\mathcal{C}| \text{Vol}_q(n, d-1).$$

Taking logarithms base q gives the claimed lower bound on k . $\square$

Remark. **Varshamov [1957]** observed that the above greedy construction can be made linear. Whenever there is such an x , replace $\mathcal{C}$ with $\text{span}(\mathcal{C} \cup \{x\})$. For every $\alpha \neq 0$ and $c \in \mathcal{C}$,

$$\text{wt}(c + \alpha x) = \Delta(-\alpha^{-1}c, x) \geq d.$$

Definition 15. For $q \geq 2$ and $0 \leq \delta \leq 1$, the q -ary entropy function is

$$H_q(\delta) = \delta \log_q(q-1) - \delta \log_q(\delta) - (1-\delta) \log_q(1-\delta),$$

where $0 \log_q(0)$ is interpreted as 0.

Proposition 16: Let $q \geq 2$ be an integer and $0 \leq \delta \leq 1 - 1/q$. Then:

- $\text{Vol}_q(n, \lfloor \delta n \rfloor) \leq q^{H_q(\delta)n}$;
- $\text{Vol}_q(n, \lfloor \delta n \rfloor) \geq q^{(H_q(\delta) - o(1))n}$.

We will not prove the above lemma and leave it as an exercise.

Theorem 17 (Asymptotic Hamming). Let $q \geq 2$ be fixed, and let $\mathcal{C} = \{C_i\}_{i \in \mathbb{N}}$ be a family of q -ary codes. Then

$$R(\mathcal{C}) \leq 1 - H_q(\delta(\mathcal{C})/2).$$

Theorem 18 (Asymptotic GV). Let $q \geq 2$ be fixed. For every $\delta \in (0, 1 - 1/q)$, there exists a family of q -ary codes $\mathcal{C} = \{C_i\}_{i \in \mathbb{N}}$ such that $\delta(\mathcal{C}) \geq \delta$ and

$$R(\mathcal{C}) \geq 1 - H_q(\delta).$$

Open Question 19. Can you construct a family of binary codes meeting the GV bound in deterministic polynomial time with positive rate and distance?

Theorem 20 (Random linear codes achieve the GV bound). Let q be a prime power, let $0 < \delta < 1 - 1/q$, and let $0 < \varepsilon < 1 - H_q(\delta)$. For sufficiently large n , set

$$k = \lfloor (1 - H_q(\delta) - \varepsilon)n \rfloor$$

and choose G uniformly at random from $\mathbb{F}_q^{k \times n}$. If $\mathcal{C} = \{mG \mid m \in \mathbb{F}_q^k\}$, then

$$\Pr_G[\mathcal{C} \text{ is an } (n, k, \lceil \delta n \rceil)_q \text{ linear code}] \geq 1 - q^{-\varepsilon n}.$$

Proof. Fix a nonzero message $m \in \mathbb{F}_q^k$. Since G is uniformly random, mG is uniformly distributed over $\mathbb{F}_q^n$. Hence

$$\Pr_G[\text{wt}(mG) < \lceil \delta n \rceil] = \frac{\text{Vol}_q(n, \lceil \delta n \rceil - 1)}{q^n} \leq q^{-(1-H_q(\delta))n}.$$

Taking a union bound over all nonzero $m \in \mathbb{F}_q^k$ gives

$$\Pr_G[\exists m \neq 0 : \text{wt}(mG) < \lceil \delta n \rceil] \leq q^k q^{-(1-H_q(\delta))n} \leq q^{-\varepsilon n}.$$

Outside this event, every nonzero message maps to a word of weight at least $\lceil \delta n \rceil$. In particular, the map $m \mapsto mG$ is injective, so G has rank k . The distance bound now follows from the linearity of the code and the q -ary version of Claim 1 since $\Delta(c, c') = \text{wt}(c - c')$. $\square$

Question 21. What happens in the regime when $\delta \geq 1 - 1/q$?

Up next: Dual codes, Hadamard Code, and Plotkin Bound:

Definition 22 (Dual Code). Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a linear code with parity-check matrix H . The *dual code* $\mathcal{C}^\perp$ is the code generated by H^T . In particular,

$$\mathcal{C}^\perp = \{x \in \mathbb{F}_q^n \mid x \cdot c = 0 \text{ for all } c \in \mathcal{C}\}.$$

References

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